



Taming the Cloud: Provisioning with Terraform

Terraform is open source software that enables sysadmins and developers to write, plan and create infrastructure as code. It is a no-frills software package, which is very simple to set up. It uses a simple configuration language or JSON, if you wish.



Terraform is a tool to create and manage infrastructure that works with various IaaS, PaaS and SaaS service providers. It is very simple to set up and use, as there aren't multiple packages, agents and servers, etc. involved. You just declare your infrastructure in a single (or multiple) file using a simple configuration language (or JSON), and that's it. Terraform takes your configurations, evaluates the various building blocks from those to create a dependency graph, and presents you a plan to create the infrastructure. When you are satisfied with the creation plan, you apply the configurations and Terraform creates independent resources in parallel. Once some infrastructure is created using Terraform, it compares the current state of the infrastructure with the declared configurations on subsequent runs, and only acts upon the changed part of the infrastructure. Essentially, it is a CRUD (Create Read Update Destroy) tool and acts on the infrastructure in an idempotent manner.

Installation and set-up

Terraform is created in Golang, and is provided as a static binary without any install dependencies. You just pick the correct binary (for GNU/Linux, Mac OS X, Windows, FreeBSD, OpenBSD and Solaris) from its download site, unzip

it anywhere in your executable's search path and all is ready to run. The following script could be used to download, unzip and verify the set-up on your GNU/Linux or Mac OS X nodes:

```
HCTLSLOC="/usr/local/bin"
HCTLSURL="https://releases.hashicorp.com/"
# use latest version shown on https://www.terraform.io/
downloads.html
TRFRMVER="x.y.z"

if uname -v | grep -i darwin 2>&1 > /dev/null
then
    OS='darwin'
else
    OS='linux'
fi

wget -P /tmp --tries=5 -q -L "${HCTLSURL}/
terraform_${TRFRMVER}/${terraform}_${TRFRMVER}_${OS}_amd64.zip"
sudo unzip -o -o "/tmp/terraform_${TRFRMVER}_${OS}_amd64.zip" -d
"${HCTLSLOC}"
rm -fv "/tmp/terraform_${TRFRMVER}_${OS}_amd64.zip"
terraform version
```